

Aurora OS shim (Qt/C++ tun)

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Volume 4 (Clients) nano-detail specification — deepens the **Aurora OS** row of the per-platform shim matrix in doc 03 [03-client §5, §5.6] into an implementation-ready spec of the **omprussia Flutter-fork + Qt/C++ + /dev/net/tun** shim that satisfies the TunnelPlatform contract [03-client §4] and hosts the Rust **helix-core** over a **C ABI** [03-client §5, D-CLIENT-3]. SPEC ONLY: describes **what to build** (signatures, channel contracts, state machines, lifecycle, memory budgets, error taxonomy, edge cases, test points) — it does **not** build the product.

Boundary. This document **consumes** the FFI surface + TunnelStatus enum frozen by doc 03 [03-client §3] and the orchestrator/status contract owned by Volume 2 [v02 orchestrator-and-state §4.1, §5]. It **owns** only the Aurora-specific seam: the Qt/C++ tunnel backend, the Friflex Flutter-plugin channel bridge, the /dev/net/tun lifecycle, the cbindgen C-ABI binding to **helix-core**, and the Aurora build/sign pipeline. Every other pixel is shared Dart and is **out of scope** here [03-client §0, §5 "the UI ports for free, the shim does not"].

Evidence base. Citations inline by id: [03-client §N] = final/03-client-core-and-ui.md; [v02-tt §N] = final/v02-data-plane/transport-trait.md; [v02-orch §N] = final/v02-data-plane/orchestrator-and-state.md; [04_ARCH §N] = 04_VPN_CLD/HelixVPN-Architecture-Refined.md; [04_UI §N] = 04_VPN_CLD/HelixVPN-helix-ui-Flutter.md; [research-flutter_ffi] = v09-research/research-flutter_ffi.md; [SYN §N] = v09-research/_SYNTHESIS.md. Claims not grounded in this evidence base are tagged UNVERIFIED per constitution §11.4.6 — never fabricated.

Table of contents

- 0. Position, ownership, and the one-sentence contract
- 1. Why Aurora is the hardest seam (the real-native-work risk)
- 2. Process & seam topology (where the core lives)
- 3. The cbindgen C-ABI binding to **helix-core**
- 4. The Friflex channel contract (Dart $\rightleftharpoons$ Qt/C++)
- 5. The Dart side — AuroraTunnelPlatform + AuroraHelixCore

- 6. The Qt/C++ backend — HelixTunnelBackend
- 7. /dev/net/tun lifecycle, routes, DNS, kill-switch
- 8. Shim lifecycle state machine
- 9. TunnelStatus consistency + the status bridge
- 10. Error taxonomy + edge cases
- 11. Memory / size / performance budgets
- 12. Build, RPM packaging, signing
- 13. Test points — tied to the §11.4.169 test-type vocabulary
- 14. Open decisions surfaced by this document
- 15. Cross-document contracts this document fixes
- Sources verified

0. Position, ownership, and the one-sentence contract

The one-sentence contract. The Aurora shim is *one TunnelPlatform implementation* [03-client §4] that, on Aurora OS, (a) configures a `/dev/net/tun` device, (b) hands its fd to a `helix-core` instance **linked as a C library into a Qt/C++ backend**, and (c) reports lifecycle + bridges the core's `TunnelStatus` back to the shared Flutter UI over **Friflex** Flutter- plugin channels — and does **nothing else** [03-client §5.6, 04_UI §6.1, 04_ARCH §5.3].

It owns exactly five things and nothing more:

#	Owned here	Consumed from
A1	The cbindgen C-ABI header <code>helix-core</code> exposes to non-Dart native hosts (Aurora is the canonical raw-C consumer, D-CLIENT-3)	<code>helix-ffi</code> crate → doc 03 owns the Rust surface
A2	The Friflex channel contract (<code>MethodChannel/EventChannel</code> names + payload schemas)	<code>TunnelPlatform/ HelixCore</code> abstractions [03-client §3, §4]
A3	The Qt/C++ HelixTunnelBackend that opens the tun, links the core, pumps lifecycle	OS: <code>/dev/net/tun</code> , <code>ConnMan</code> , <code>Sailjail</code> (Aurora primitives)
A4	The <code>/dev/net/tun</code> + routes + DNS + kill-switch Aurora mechanics	<code>TunnelConfig</code> [03-client §4]
A5	The Aurora RPM build/sign pipeline	<code>melos</code> build matrix [03-client §11]

It does **NOT** own: any screen/widget/state (shared Dart, [03-client §7-§8]); the `helix-core` internals, the `Transport` trait, the orchestrator loops ([v02-tt], [v02-orch]); the `TunnelStatus` *enum definition* (frozen by [v02-orch §4.1], mirrored by [03-client §3.1]); or the control-plane wire ([02] doc).

0.1 Aurora-specific invariants (extend the §0.1 client invariants)

#	Invariant	Source
AI1	On Aurora the core lives in the Qt/C++ backend , not the Flutter process — therefore <code>statusStream</code> ,	

#	Invariant	Source
	setShields, setExit, exits, advertise all cross the Friflex channel (not frb), because frb-in-the-embedder is UNVERIFIED on the omprussia fork (§2, §14 D-AUR-1).	[03-client §5.6], [research-flutter_ffi §4]
AI2	The shim hands the core a tun fd (attach_tun(fd) model, like Linux/Android) — it never does crypto/obfuscation; helix-core is the only place WG + transport live (O2 [03-client §4.1]).	[03-client §4.1 O2], [v02-tt I1]
AI3	Aurora is an enterprise/government SKU with Russian-hosted toolchain (GitLab omprussia / Mos.Hub), its own CI runners + RPM signing, isolated so a fork lag never blocks mainline (§12).	[03-client §5.6], [04_ARCH §5.7]
AI4	Every Aurora-required OS API the embedder/SDK does not expose is an honest SKIP-with-reason (§11.4.3), never a faked PASS; Phase-3 biggest-platform-risk discipline [SYN §4].	[research-flutter_ffi §4], [03-client §13 T3.2]
AI5	events() reports revoked when ConnMan / the OS / an admin (device.revoked, doc 02) kills the tunnel out-of-band; the UI shows it within budget, never claims "connected" (O3, CI2).	[03-client §4.1 O3]

1. Why Aurora is the hardest seam (the real-native-work risk)

Aurora OS (Sailfish-derived, Qt/QML-native, Linux-kernel-based) has **no mainline Flutter support**; the only single-codebase path is the **OMP Russia fork** `gitlab.com/omprussia/flutter` (flutter SDK fork + flutter-embedder runtime + flutter-community-plugins group, plugins follow the *_aurora naming pattern) [research-flutter_ffi §4, 04_ARCH §5.2]. Phase-3 honest risk register [SYN §4, 03-client §13]:

Risk	Concrete shape	Mitigation in this spec
R1 Forked, lagging Flutter	flutter-aurora tracks a Flutter baseline behind mainline; plugins need Aurora variants	Pin the exact fork tag; isolate the Aurora runner (§12); the UI ports unchanged, only this shim is bespoke (AI3)
R2 No frb-in-embedder guarantee	dart:ffi DynamicLibrary loading a cdylib on the omprussia embedder is UNVERIFIED [research-flutter_ffi §4]	Put the core in the Qt/C++ backend via cbindgen C ABI (§3) — depends only on a C ABI (universal) + Friflex MethodChannel (provided), not on frb (AI1, D-AUR-1)
R3 VPN-permission / tun API unknown	Whether a sandboxed (Sailjail) Aurora app may open("/dev/net/tun") + hold CAP_NET_ADMIN is UNVERIFIED (§7.1)	Two paths: in-process raw tun if permitted, else a privileged D-Bus helper daemon (§7.2); honest SKIP if neither is reachable (AI4)

Risk	Concrete shape	Mitigation in this spec
R4 Route/DNS via ConnMan	Aurora uses ConnMan for connectivity; route+DNS install is ConnMan/netlink, not Android VpnService.Builder	Spec the ConnMan + netlink path; mark ConnMan D-Bus specifics UNVERIFIED (§7.3)
R5 No device in lab	The team likely lacks an Aurora device/emulator for autonomous device tests	Device-gated test rows are honest SKIP-with-reason per §11.4.3/§11.4.52 (§13), never a bluff PASS

The payoff is still real [03-client §5 "the UI ports for free"]:
ConnectButton, ExitPicker, StatusChip, settings, account, policy views — **every pixel** — are the identical shared Dart. Adding Aurora is "write **this** shim + bend the build," not "port the app."

2. Process & seam topology (where the core lives)

On most platforms the core is loaded into (or beside) the Flutter process and status flows core → frb → UI [03-client §3.3]. On Aurora the core lives in the **Qt/C++ backend** [03-client §5.6], so **both** the lifecycle commands **and** the status/logic calls cross a **Friflex channel**:

```

flowchart TB
    subgraph FlutterProc ["Flutter process (omprussia flutter-embedder)"]
        UI["Shared Flutter UI (Dart)<br/>ConnectButton · StatusChip · ExitPicker · Shields"]
        ATP["AuroraTunnelPlatform (Dart)<br/>impl TunnelPlatform (§5.1)"]
        AHC["AuroraHelixCore (Dart)<br/>impl HelixCore over channels (§5.2)"]
        UI -->|"startTunnel / stopTunnel"| ATP
        UI -->|"statusStream / setShields / setExit / exits"| AHC
    end
    subgraph FriflexPlugin ["Friflex Flutter plugin (registers channels)"]
        MC["MethodChannel 'helixvpn/tunnel'<br/>+ 'helixvpn/core'"]
        EC1["EventChannel 'helixvpn/tunnel/events'<br/>>(PlatformTunnelEvent)"]
        EC2["EventChannel 'helixvpn/core/status'<br/>>(TunnelStatus JSON)"]
    end
    subgraph QtBackend ["Qt/C++ backend process – HelixTunnelBackend (§6)"]
        HTB["HelixTunnelBackend (QObject)"]
        TUN["/dev/net/tun fd + ConnMan routes/DNS (§7)"]
        CORE["helix-core (Rust)<br/>linked as C lib via cbindgen (§3)"]
    end

```

```

HTB -->|"open/configure"| TUN
HTB -->|"helix_core_start + attach_tun(fd)"| CORE
CORE -->|"HxStatusCb(json)"| HTB
TUN <-->|"packets (in-process, I1 opaque)"| CORE
end
ATP <-->|"invoke / result"| MC
AHC <-->|"invoke / result"| MC
EC1 -->|"up/down/permissionDenied/revoked/error"| ATP
EC2 -->|"Connecting...Connected...Down"| AHC
MC <--> HTB
EC1 <--> HTB
EC2 <--> HTB

```

UNVERIFIED (§2.1): whether the Friflex plugin runs the Qt backend **in-process** with the embedder (same address space, fd shared directly) or as a **separate D-Bus service** (fd passed via SCM_RIGHTS over a Unix socket) is an Aurora-platform implementation detail not settled by the evidence base. The contract below (§3–§6) is identical either way; §7.2 covers the separate-service fd-passing case. Phase-0/3 spike settles it (D-AUR-2).

Why core-in-Qt-backend, not core-in-Flutter-via-frb (A11, R2): the C ABI (cbindgen) is universal and proven; frb on the omprussia embedder is unproven. Hosting the core in the Qt/C++ backend trades a small duplication (the status plumbing is a channel instead of frb) for **eliminating the frb-on-fork unknown** — the right Phase-3 risk posture. The alternative (frb in the embedder) is surfaced as **D-AUR-1** (§14) and gated on a spike.

3. The cbindgen C-ABI binding to helix-core

helix-core exposes **two** generator-fed surfaces from one Rust source [03-client §3, D-CLIENT-3]: **frb** for Dart (mainline platforms) and a **cbindgen C header** for native hosts where frb/UniFFI are thin — Aurora (Qt/C++) and HarmonyOS (NAPI) are the raw-C consumers. This section pins the C ABI the Aurora backend links against.

3.1 The C header (helix_core.h, cbindgen-generated)

```

/* helix_core.h – generated by cbindgen from helix-ffi/src/
   c_abi.rs.
 * Hand-authored Rust source is helix-ffi/src/api.rs (frb) [03-
   client §3.1];
 * c_abi.rs is a thin #[no_mangle] extern "C" wrapper over the
   SAME functions,
 * so frb-Dart and C-Aurora drive byte-identical core logic (no
   fork). */
#ifdef HELIX_CORE_H
#define HELIX_CORE_H
#include <stdint.h>

/* ---- return codes (mirror anyhow::Result -> i32; 0 == Ok) ----
 */

```

```

typedef enum HxResult {
    HX_OK                = 0,
    HX_ERR_CONFIG        = 1,    /* bad/empty token, unknown
        transport string */
    HX_ERR_TUN           = 2,    /* attach_tun: fd invalid / not a
        tun device */
    HX_ERR_ALREADY       = 3,    /* start called while already
        running */
    HX_ERR_NOT_RUNNING   = 4,    /* stop/detach/logic call while
        stopped */
    HX_ERR_AUTH          = 5,    /* WG handshake rejected (revoked/
        bad key) */
    HX_ERR_INTERNAL      = 6,    /* unexpected; detail in last-error
        string */
} HxResult;

typedef enum HxMode { HX_MODE_CLIENT = 0, HX_MODE_CONNECTOR = 1 }
HxMode;

/* ---- lifecycle (logic only – NOT the OS tunnel; the shim owns
    tun, AI2) ---- */
/* start: spins up the core's tokio runtime + orchestrator (v02-
    orch §2),
    * arms the kill-switch (v02-orch §8.2), emits Connecting.
    BLOCKING but bounded:
    * returns once the orchestrator reaches Connecting (sub-ms);
    status thereafter
    * arrives via the registered callback (§3.2). */
HxResult helix_core_start(const char* session_or_map_token,
                        const char* transport /*
                            "auto"|"plain"|"masque"|... */
                        HxMode mode);

/* attach_tun: hand the core the tun fd; the core OWNS the fd
    lifetime until
    * detach_tun/stop. Must be called AFTER start, BEFORE the WG
    handshake can pump.
    * Linux/Aurora tun fd model [03-client §5 matrix]. */
HxResult helix_core_attach_tun(int32_t fd);
HxResult helix_core_detach_tun(void);

/* stop: graceful teardown (v02-orch §2.4): close transport,
    revert DNS,
    * kill-switch per §8.6, join loops within shutdown_grace, emit
    Down{stopped}. */
HxResult helix_core_stop(void);

/* ---- logic calls (no OS tunnel touched; UI->channel->backend-
    >core) ---- */
HxResult helix_core_set_shields(const char* shields_json); /*
    schema §4.4 */
HxResult helix_core_set_exit(const char* exit_id,
                            const char* multihop_chain_json /*
                                nullable */);
/* exits/advertise return owned UTF-8 JSON; caller MUST
    helix_core_string_free.

```

```

    * On error return NULL and set last-error
      (helix_core_last_error). */
char* helix_core_exits_json(void); /* JSON
array, schema §4.5 */
char* helix_core_advertise_json(const char* cidrs_json); /*
connector mode */
void helix_core_string_free(char* s);

/* ---- status push (core -> backend), the statusStream source on
Aurora ---- */
/* cb is invoked from a core tokio thread; the backend MUST
marshal to the Qt
* event loop before touching Friflex (§6.3). json schema = §9.1.
user is opaque. */
typedef void (*HxStatusCb)(const char* status_json, void* user);
void helix_core_set_status_cb(HxStatusCb cb, void* user);

/* ---- diagnostics (redacted; never secrets, §11.4.10) ---- */
const char* helix_core_last_error(void); /* thread-local, valid
until next core call */
const char* helix_core_version(void); /* "helixvpn-<ver>"
§11.4.151 */

#endif /* HELIX_CORE_H */

```

3.2 C-ABI $\rightleftharpoons$ frb correspondence (no fork, I4)

Each C function is the synchronous facade over the identical frb async surface [03-client §3.1]; cbindgen cannot express async fn, so the wrapper blocks on the core's internal tokio runtime (bounded) and pushes continuous status via the HxStatusCb callback (the C analogue of frb's StreamSink<TunnelStatus> [research-flutter_ffi §1]):

frb (Dart) [03-client §3.1]	C ABI (Aurora)	Notes
start(cfg) -> Result<()>	helix_core_start(token, transport, mode)	cfg fields flattened to C args
stop() -> Result<()>	helix_core_stop()	idempotent
status_stream(sink: StreamSink<TunnelStatus>)	helix_core_set_status_cb(cb, user)	callback $\cong$ StreamSink
exits() -> Result<Vec<ExitOption>>	helix_core_exits_json() $\rightarrow$ free	JSON-serialized (§4.5)
set_exit(id, chain)	helix_core_set_exit(id, chain_json)	
set_shields(Shields)	helix_core_set_shields(json)	(§4.4)
advertise(cidrs)	helix_core_advertise_json(cidrs_json)	connector only
attach_tun(fd) / detach_tun()	helix_core_attach_tun(fd) / helix_core_detach_tun()	tun fd model (AI2)

Anti-bluff (§11.4.27/§11.4.50). The frb path and the C-ABI path **MUST** be proven equivalent by a **shared serialization test-vector corpus** (the same pattern the edge uses for encode/decode [v02-tt §8.3]): a frozen set of (core-action, status-JSON) and (exits-call, exits-JSON) pairs both the frb Dart binding and the C-ABI consumer must satisfy bit-for-bit. A divergence is a release blocker — the mechanical proof that "Aurora drives the same core."

3.3 cbindgen build configuration

```
# helix-ffi/cbindgen.toml (the C-ABI half of D-CLIENT-3)
language = "C"
include_guard = "HELIX_CORE_H"
[export]
include = ["HxResult", "HxMode", "HxStatusCb"]
prefix = "helix_core_" # symbol namespace; matches §11.4.151 project prefix
[fn]
sort_by = "None"
[parse]
parse_deps = false # only helix-ffi's own
                #[no_mangle] surface
```

helix-ffi builds for Aurora as `crate-type = ["staticlib"]` (linked into the Qt backend, smallest footprint) targeting the Aurora ARM triple (UNVERIFIED exact triple — Aurora is ARM/aarch64; settled by the §12 SDK), with `opt-level = "z"` + LTO + `panic = "abort"` + `strip` (the lean-core recipe [03-client §10, §5.1]). The Aurora build does **not** carry the iOS NE 15 MB ceiling (Aurora is a full Linux process, §11), but stays lean by AI3 discipline.

4. The Friflex channel contract (Dart $\rightleftharpoons$ Qt/C++)

Three channels, registered by the Friflex Flutter plugin on Aurora [03-client §5.6, 04_UI §6.1]. Two are the standard `TunnelPlatform` channels [03-client §4]; one is the **Aurora-specific status bridge** (AI1, because the core is not in the Flutter process).

Channel	Type	Direction	Carries
helixvpn/ tunnel	MethodChannel	Dart → backend	startTunnel, stopTunnel (the TunnelPlatform verbs)
helixvpn/ tunnel/ events	EventChannel	backend → Dart	PlatformTunnelEvent (lifecycle only, [03-client §4])
helixvpn/ core	MethodChannel	Dart → backend	setShields, setExit, exits, advertise (the HelixCore logic verbs)
helixvpn/ core/status	EventChannel	backend → Dart	TunnelStatus JSON (the statusStream source, §9)

4.1 helixvpn/tunnel method payloads

```
// invoke("startTunnel", <TunnelConfig>) – TunnelConfig per [03-client §4]
{ "overlayIp": "100.64.0.7/32", "routes": ["0.0.0.0/0"],
  "dnsServers": ["100.64.0.1"], "splitExcludeApps": [],
  "mtu": 1280, "sessionOrMapToken": "<opaque session/map token>" }
// invoke("stopTunnel") – no args; returns null on success,
PlatformException on failure
```

startTunnel returns null (success) once the backend has reached the **Up** state (§8) — i.e. tun created + core started + first Connected/Handshaking observed — OR throws a PlatformException(code, detail) mapping to a PlatformTunnelEvent kind (§10). It is **idempotent** (O1 [03-client §4.1]): a second startTunnel while Up is a no-op success.

4.2 helixvpn/tunnel/events payloads (lifecycle, [03-client §4])

```
{ "kind": "up" } // PlatformTunnelEventKind.up
{ "kind": "down" } // graceful or post-stop
{ "kind": "permissionDenied" } // tun/ CAP_NET_ADMIN refused (§7.1)
{ "kind": "revoked", "detail": "connman-off" } // OS/admin killed tunnel (AI5/03)
{ "kind": "error", "detail": "tun open EPERM" } // honest reason (§11.4.6 no guessing)
```

4.3 helixvpn/core method payloads

```
invoke("setShields", <Shields JSON §4.4>) -> null | PlatformException
invoke("setExit", { "id": "se-fra-1", "multiHopChain": ["se-fra-1", "se-zrh-2"] }) -> null
invoke("exits") -> [ <ExitOption §4.5>, ... ]
invoke("advertise", { "cidrs": ["10.10.0.0/24"] }) -> { "accepted": [...], "conflicts": [...] }
```

4.4 Shields JSON (mirrors Shields [03-client §3.1])

```
{ "killSwitch": true, "dnsProtection": true, "daita": false,
  "postQuantum": false, "splitTunnel": ["10.0.0.0/8"] }
```

4.5 ExitOption JSON (mirrors ExitOption [03-client §3.1])

```
{ "id": "se-fra-1", "kind": "privacy_exit", "label": "Frankfurt",
  "country": "DE", "rttMs": 23, "jurisdiction": "EU" }
```

4.6 Channel codec + threading rules

- All payloads are the **standard Flutter StandardMessageCodec** maps/lists (the Friflex plugin marshals to/from the Qt QVariantMap). JSON-shaped above for clarity; the wire type is the codec's native map.
 - The backend **MUST never** block the Qt event loop in a channel handler (§6.3): startTunnel/stopTunnel dispatch to a worker and complete the MethodChannel.Result asynchronously.
 - EventChannel sinks are touched **only** from the Qt main thread; the core's HxStatusCb (a tokio thread) is marshalled via QMetaObject::invokeMethod(..., Qt::QueuedConnection) before any sink call (§6.3) — a cross-thread sink write is a defect.
 - No SecretBytes-class value (session token aside, which is opaque) crosses any channel as a logged field; channel tracing redacts sessionOrMapToken (§11.4.10).
-

5. The Dart side — AuroraTunnelPlatform + AuroraHelixCore

The shared Dart UI is **unchanged**; Aurora provides two channel-backed implementations of the existing abstractions [03-client §3.2, §4] selected by the platform at runHelixApp wiring time [03-client §6]. No widget knows it is on Aurora (CI2 honesty preserved).

5.1 AuroraTunnelPlatform (impl TunnelPlatform)

```
// apps/access/aurora/lib/aurora_tunnel_platform.dart
class AuroraTunnelPlatform implements TunnelPlatform {
  static const _m = MethodChannel('helixvpn/tunnel');
  static const _e = EventChannel('helixvpn/tunnel/events');

  @override
  Future<void> startTunnel(TunnelConfig cfg) async {
    try {
      await _m.invokeMethod<void>('startTunnel',
        cfg.toMap()); // §4.1
    } on PlatformException catch (pe) {
      // map to the honest event; do NOT swallow (01: leave no
      // half-open tun)
      throw TunnelStartException(_kindFromCode(pe.code),
        pe.message);
    }
  }

  @override
  Future<void> stopTunnel() =>
    _m.invokeMethod<void>('stopTunnel');

  @override
  Stream<PlatformTunnelEvent> events() => _e
```

```

        .receiveBroadcastStream()
        .map((m) => PlatformTunnelEvent(
            _kind(m['kind'] as String), m['detail'] as String?));
    }

```

5.2 AuroraHelixCore (impl HelixCore — the Aurora status/logic bridge)

Because the core lives in the Qt backend (AI1), Aurora's HelixCore is **channel-backed** instead of frb-backed — the **only** platform where this is true. The Dart TunnelStatus it yields is the **same generated/mirrored type** [03-client §3.1], deserialized from the status JSON (§9.1), so every widget folds it identically (§8 of doc 03).

```

// apps/access/aurora/lib/aurora_helix_core.dart
class AuroraHelixCore implements HelixCore {
    static const _m = MethodChannel('helixvpn/core');
    static const _status = EventChannel('helixvpn/core/status');

    @override
    Future<void> start({required String transport, String?
        mapPathOrSession,
        CoreMode mode = CoreMode.client}) =>
        // on Aurora, start is driven THROUGH the tunnel shim
        // (startTunnel) because
        // the tun fd must exist before the core pumps; this is a
        // no-op pass-through
        // kept for interface symmetry. The real start is
        // AuroraTunnelPlatform.startTunnel.
        Future<void>.value();

    @override
    Future<void> stop() => Future<void>.value(); // ditto ->
        stopTunnel

    @override
    Stream<TunnelStatus> statusStream() => _status
        .receiveBroadcastStream()
        .map((j) => TunnelStatusJson.decode(j as
            Map)); // §9.1 -> mirrored enum

    @override
    Future<List<ExitOption>> exits() async {
        final raw = await _m.invokeMethod<List<dynamic>>('exits');
        return raw!.map((e) => ExitOption.fromMap(e as Map)).toList();
    }

    @override
    Future<void> setExit(String id, {List<String>? multiHopChain})
        =>
        _m.invokeMethod<void>('setExit', {'id': id,
            'multiHopChain': multiHopChain});

```

```

@override
Future<void> setShields(Shields s) =>
    _m.invokeMethod<void>('setShields', s.toMap());

@override
Future<AdvertiseResult> advertise(List<String> cidrs) async =>
    AdvertiseResult.fromMap(
        (await _m.invokeMethod<Map>('advertise', {'cidrs':
            cidrs})))!);

@override
Future<void> attachTun(int fd) => Future<void>.value(); //
    backend-internal on Aurora

@override
Future<void> detachTun() => Future<void>.value();
}

```

5.3 Wiring (the only Aurora-aware line in helix_domain)

```

// helix_domain/lib/run_helix_app.dart – platform select (branch
    on the FORK target,
// the one permitted Platform check; UI never branches on
    Platform.isX, [03-client §7.3])
helixCoreProvider.overrideWith((ref) =>
    capabilities.contains(Capability.tunnel)
        ? (HelixPlatform.isAurora ? AuroraHelixCore() :
            RealHelixCore() /* frb */)
        : NoCore());
tunnelPlatformProvider.overrideWith((ref) =>
    HelixPlatform.isAurora ?
        AuroraTunnelPlatform() : /* per-OS shim */ defaultShim());

```

HelixPlatform.isAurora is a **compile-time** flavor constant set by the Aurora build (a `--dart-define=HELIX_TARGET=aurora`), not a runtime probe — so the non-Aurora branches tree-shake out of every other build (lean binary, §6 of doc 03).

6. The Qt/C++ backend — HelixTunnelBackend

The Qt/C++ class that the Friflex plugin instantiates; it is the native owner of the tun, the core, and the channel handlers.

6.1 Class sketch

```

// shims/aurora/HelixTunnelBackend.h
#include <QObject>
#include <QVariantMap>
#include <atomic>
extern "C" {
    #include "helix_core.h"           // §3.1 cbindgen header
}

```

```

class HelixTunnelBackend : public QObject {
    Q_OBJECT
public:
    explicit HelixTunnelBackend(QObject* parent = nullptr);
    ~HelixTunnelBackend() override;    // RAI: stopTunnel() on
        destroy (04 cleanup)

    // MethodChannel 'helixvpn/tunnel' handlers (async; complete the
        Result later)
    void startTunnel(const QVariantMap& cfg, MethodResult
        result);    // §4.1
    void stopTunnel(MethodResult result);

    // MethodChannel 'helixvpn/core' handlers
    void setShields(const QVariantMap& shields, MethodResult
        result);
    void setExit(const QString& id, const QVariantList& chain,
        MethodResult result);
    void exits(MethodResult result);
    void advertise(const QStringList& cidrs, MethodResult result);

signals:
    // marshalled to the EventChannels on the Qt main thread (§6.3)
    void lifecycleEvent(const QVariantMap& event);    // ->
        'helixvpn/tunnel/events'
    void coreStatus(const QString& statusJson);    // ->
        'helixvpn/core/status'

private:
    // C status callback trampoline: tokio-thread -> Qt main thread
        (§6.3)
    static void onStatus(const char* json, void* user);

    int tunFd_ {-1};    // /dev/net/tun fd
        (§7)
    std::atomic<int> state_ {Idle};    // ShimState (§8);
        atomic for cross-thread reads
    QString ifName_;    // e.g. "helix0"
    ConnmanGuard connman_;    // RAI
        routes+DNS+kill-switch (§7.3)
};

```

6.2 startTunnel flow (the make-it-work path)

```

void HelixTunnelBackend::startTunnel(const QVariantMap& cfg,
    MethodResult result) {
    if (state_ == Up) { result.success(); return; }    // 01
        idempotent
    QtConcurrent::run([this, cfg, result]() mutable {
        setState(RequestingPermission);
        if (!ensureVpnPermission()) {    //
            §7.1 (UNVERIFIED API)
            emitLifecycle("permissionDenied");
        }
    });
}

```

```

        result.error("permissionDenied", "tun/CAP_NET_ADMIN
            refused", {});
        setState(Down); return;
    }
    setState(CreatingTun);
    tunFd_ = openTun(&ifName_); //
        §7.2: ioctl(TUNSETIFF)
    if (tunFd_ < 0) {
        emitLifecycle("error", strerror(errno));
        result.error("tunOpen", QString("tun open:
            %1").arg(strerror(errno)), {});
        setState(Down); return;
    }
    if (!connman_.apply(ifName_, cfg)) { //
        §7.3 routes+DNS+kill-switch
        ::close(tunFd_); tunFd_ = -1;
        emitLifecycle("error", "route/dns apply failed");
        result.error("netConfig", "ConnMan/netlink apply failed",
            {});
        setState(Down); return;
    }
    setState(CoreStarting);
    helix_core_set_status_cb(&HelixTunnelBackend::onStatus, this);
    HxResult r =
        helix_core_start(cfg["sessionOrMapToken"].toString().toUtf8(),
            "auto", HX_MODE_CLIENT); //
        transport from policy
    if (r == HX_OK) r = helix_core_attach_tun(tunFd_); //
        hand fd to core (AI2)
    if (r != HX_OK) {
        teardown(/*userInitiated=*/false); // 04
        cleanup on every exit path
        emitLifecycle("error", helix_core_last_error());
        result.error("coreStart", QString::number(r), {});
        setState(Down); return;
    }
    setState(Up);
    emitLifecycle("up");
    result.success(); //
        §4.1 returns once Up
    });
}

```

6.3 Status trampoline (the cross-thread rule)

```

// invoked from a core tokio thread – MUST NOT touch Friflex/Qt
// sinks directly
void HelixTunnelBackend::onStatus(const char* json, void* user) {
    auto* self = static_cast<HelixTunnelBackend*>(user);
    const QString s =
        QString::fromUtf8(json); // copy out before
        returning
    QMetaObject::invokeMethod(self, [self, s]() {

```

```

        emit self->coreStatus(s); // now
        on Qt main thread
    }, Qt::QueuedConnection);
}

```

The copy-out-before-return is mandatory: json is owned by the core and valid only for the callback duration (the C analogue of frb's StreamSink lifetime). A torn read or a sink touch from the tokio thread is a defect caught by the RACE test (§13).

6.4 stopTunnel / teardown (O4 — quiescent on every exit path)

```

void HelixTunnelBackend::teardown(bool userInitiated) {
    helix_core_detach_tun();
    helix_core_stop(); // v02-orch §2.4
    graceful order
    if (tunFd_ >= 0) { ::close(tunFd_); tunFd_ = -1; }
    connman_.revert(userInitiated); // §8.6: kill-switch
    reverts ONLY on user stop
    emitLifecycle("down");
}

```

~HelixTunnelBackend() calls teardown(false) (RAII) so a backend crash leaves the kill-switch **closed** (O-I11 [v02-orch §8.4], no leak on abnormal exit), and no orphan tun/routes/DNS survive (O4 [03-client §4.1]).

7. /dev/net/tun lifecycle, routes, DNS, kill-switch

7.1 VPN permission (the UNVERIFIED gate, R3)

Whether a Sailjail-sandboxed Aurora app may open /dev/net/tun and hold CAP_NET_ADMIN is **UNVERIFIED** [research-flutter_ffi §4 "VPN permission... needs an Aurora-specific implementation, with honest SKIP where the embedder lacks a needed API"]. Two designed paths, chosen by what the platform actually grants (settled by the §13 IT/device spike, AI4):

- **P-A (in-process):** the app declares the tun/CAP_NET_ADMIN capability in its RPM manifest / Sailjail profile and opens the tun directly in HelixTunnelBackend. Lowest complexity.
- **P-B (privileged helper):** a small systemd/D-Bus-activated helper daemon (running with CAP_NET_ADMIN) opens the tun and passes the fd to the unprivileged backend via SCM_RIGHTS over a Unix socket (§7.4). Required if Sailjail denies P-A. Mirrors the Windows privileged-service pattern [03-client §5.3] and the Linux polkit-helper note [03-client §5.4].

ensureVpnPermission() returns false ⇒ permissionDenied event (not error, O1) ⇒ honest UX, no half-open tun. If **neither** path is grantable on a given Aurora build, the feature is an honest **SKIP-with-reason** (§11.4.3, AI4) — a tracked migration item, never a faked PASS.

7.2 Opening the tun (openTun)

```
// shims/aurora/tun.cpp – standard Linux tun (Aurora is Linux-
// kernel-based)
int openTun(QString* ifNameOut) {
    int fd = ::open("/dev/net/tun", O_RDWR | O_CLOEXEC);
    if (fd < 0) return -1;
    struct ifreq ifr {}; ifr.ifr_flags = IFF_TUN |
        IFF_NO_PI; // L3, no packet-info hdr
    std::strncpy(ifr.ifr_name, "helix0", IFNAMSIZ);
    if (::ioctl(fd, TUNSETIFF, &ifr) < 0) { ::close(fd); return
        -1; }
    *ifNameOut = QString::fromUtf8(ifr.ifr_name);
    // non-blocking: the core's loop owns readiness (v02-orch §3);
    // set O_NONBLOCK
    ::fcntl(fd, F_SETFL, ::fcntl(fd, F_GETFL) | O_NONBLOCK);
    return fd;
}
```

IFF_NO_PI matters: the core's helix-tun expects raw L3 packets with **no** 4-byte packet-info prefix — the same fd contract the Linux shim uses [03-client §5.4]. A wrong flag silently corrupts every packet → caught by the E2E round-trip (§13), never shipped.

7.3 Routes + DNS + kill-switch (ConnmanGuard)

Aurora uses **ConnMan** for connectivity [R4]; route/DNS install is via ConnMan's D-Bus API and/or direct rtnetlink. The core owns the kill-switch *state machine* [v02-orch §8]; the shim is the OS-effect arm that ConnMan/netlink executes.

```
// ConnmanGuard: RAII over routes + DNS + kill-switch firewall
// rules (04).
struct ConnmanGuard {
    bool apply(const QString& ifName, const QVariantMap& cfg); //
    // overlayIp, routes, dns
    void revert(bool userInitiated); //
    // §8.6 semantics
    // apply():
    // 1. ip addr add <overlayIp> dev helix0; ip link set helix0
    // up; set mtu cfg.mtu
    // 2. for r in cfg.routes: ip route add <r> dev helix0
    // (AllowedIPs)
    // 3. DNS: push cfg.dnsServers via ConnMan service config
    // (UNVERIFIED exact D-Bus
    // method name – connman.Service SetProperty
    // "Nameservers.Configuration?");
    // fallback /etc/resolv.conf rewrite if ConnMan path
    // unavailable [v02-orch §8.3]
    // 4. kill-switch: nftables default-drop egress except gateway
    // endpoints + DHCP/ND
    // (the v02-orch §8.2 Strict ruleset, executed here on
    // netlink/nft)
};
```


UNVERIFIED (§7.3): the precise ConnMan D-Bus surface for setting per-service nameservers and for a default-route override on Aurora is not in the evidence base. The netlink (ip route/ip addr) + nftables path is standard Linux and the safe fallback; the ConnMan-native path is preferred if it exists (it avoids fighting ConnMan's own DNS management). Settled by the §13 device spike.

7.4 fd-passing (P-B only)

If P-B (privileged helper) is required, the helper opens the tun and the backend receives the fd via recvmsg + SCM_RIGHTS:

```
int recvTunFd(int unixSock)
{
    // helper -> backend fd handoff
    char ctl[MSG_SPACE(sizeof(int))] = {};
    struct msghdr msg {}; struct iovec io {(void*)"x", 1};
    msg.msg_iov = &io; msg.msg_iovlen = 1; msg.msg_control = ctl;
    msg.msg_controllen = sizeof ctl;
    if (::recvmsg(unixSock, &msg, 0) <= 0) return -1;
    struct cmsghdr* c = CMSG_FIRSTHDR(&msg);
    if (!c || c->cmsg_type != SCM_RIGHTS) return -1;
    int fd; std::memcpy(&fd, CMSG_DATA(c), sizeof fd); return
    fd; // pass to attach_tun
}
```

8. Shim lifecycle state machine

The shim's own state (distinct from the core's OrchState [v02-orch §5] and the user-facing TunnelStatus [v02-orch §4.1]). The shim state governs tun + channel lifecycle; the core state governs tunnel liveness. They are coupled at two points: CoreStarting→Up waits for the first non-Connecting core status, and a core Down{auth-failed}/OS revoke drives the shim to TearingDown.

```
stateDiagram-v2
    [*] --> Idle
    Idle --> RequestingPermission : startTunnel (§6.2)
    RequestingPermission --> Down : permissionDenied (event, §7.1)
    RequestingPermission --> CreatingTun : permission granted
    CreatingTun --> Down : tun open / route / dns failed (error
event, 04 cleanup)
    CreatingTun --> CoreStarting : tun fd + ConnmanGuard.apply OK
    CoreStarting --> Down : helix_core_start/attach_tun != HX_OK
(teardown, 04)
    CoreStarting --> Up : core status reaches Handshaking/Connected
(emit 'up')
    Up --> Up : core status flows (Connected rtt drift,
Reconnecting) over EC2
    Up --> TearingDown : stopTunnel (user) OR core Down{auth-failed}
OR ConnMan revoke
    TearingDown --> Down : detach+stop+close+connman.revert done
(emit 'down')
```

Down --> [*]
Down --> RequestingPermission : startTunnel again (fresh attempt)

8.1 Coupling table (shim transition → OS/core effect)

Shim transition	Effect	Source
Idle → RequestingPermission	check Sailjail/CAP_NET_ADMIN (§7.1)	AI4
CreatingTun	openTun + ConnmanGuard.apply (routes/DNS) + arm kill-switch closed	[v02-orch §8.2]
CoreStarting → Up	helix_core_start + attach_tun(fd); wait first non-Connecting status	AI2, §9
Up (steady)	bridge TunnelStatus over helixvpn/core/status (§9)	AI1
Up → TearingDown (user)	teardown(true) → kill-switch reverts open (§8.6)	[v02-orch §8.6]
Up → TearingDown (revoke/auth-fail/crash)	teardown(false) → kill-switch stays closed (O-I11)	[v02-orch §8.4]

8.2 Revoke / out-of-band drop (AI5/O3)

A ConnMan "connection lost" signal, a device.revoked propagated as a core Down{reason:"auth-failed"} (over EC2), or the OS killing the backend all drive Up → TearingDown and emit {"kind":"revoked"} (with an honest detail, §11.4.6). The UI must show this within the convergence budget and never paint green — the shim never claims "connected" once the core/OS says otherwise (CI2, O3). The < 3 s recovery target for a transient flap is the core's reconnect job [v02-orch §7.1], not the shim's — the shim keeps the tun + closed kill-switch in place while the core re-dials.

9. TunnelStatus consistency + the status bridge

The status enum is **frozen by the orchestrator** [v02-orch §4.1] and ****mirrored**

- extended by the client FFI** [03-client §3.1]. The Aurora status JSON is the serialization of that enum; the Dart side deserializes into the **same mirrored Dart TunnelStatus** every other platform uses — keeping "every pixel identical Dart" true on Aurora.

9.1 Status JSON schema (the helixvpn/core/status payload)

```
// serialization of the core TunnelStatus (v02-orch §4.1) extended
per [03-client §3.1].
// "tag" is the variant; payload fields are variant-specific.
{ "tag":
  "disconnected" } //
clean idle (03-client ext)
{ "tag":
  "connecting" } //
v02-orch Connecting
{ "tag":
  "handshaking" } //
v02-orch Handshaking
{ "tag": "connected", "transport": "masque-h3", "path": "direct",
  "rttMs": 23 } // +path (03-client ext)
{ "tag":
  "reconnecting" } //
v02-orch Reconnecting
{ "tag": "down", "reason": "ladder-
exhausted" } // v02-orch Down{reason}
{ "tag": "danger", "kind":
  "killswitch_tripped" } // 03-client ext (leak|
killswitch_tripped)
```

9.2 Core-enum → Aurora-JSON → Dart-mirror mapping (consistency proof)

Core OrchState/TunnelStatus [v02-orch §4.1, §5.1]	Aurora tag	Dart TunnelStatus [03-client §3.1]
Idle (pre-start) / clean post-stop	disconnected	Disconnected
Connecting	connecting	Connecting
Handshaking	handshaking	Handshaking
Connected{transport, rtt_ms} (+path from Connected.path)	connected	Connected{transport, path, rtt_ms}
Reconnecting{since_drop}	reconnecting	Reconnecting
Down{reason} (incl. auth-failed, ladder-exhausted, stopped)	down	Down{reason}
derived from Down{leak}/kill-switch trip	danger	Danger{kind}

ShuttingDown and Idle have **no** public projection ([v02-orch §5.1]): they map to connecting/disconnected at this boundary — identical to the frb projection [03-client §3.1], so Aurora is byte-consistent with every other platform. The **FFI contract test** [03-client §12] asserts the Aurora JSON round-trips to the same Dart TunnelStatus the frb path produces (the shared test-vector corpus, §3.2) — a drift FAILs the build.

9.3 Status bridge sequence

sequenceDiagram

```
participant CORE as helix-core (Rust, in Qt backend)
participant CB as HxStatusCb (tokio thread)
participant BE as HelixTunnelBackend (Qt main)
participant EC2 as EventChannel helixvpn/core/status
participant AHC as AuroraHelixCore (Dart)
participant UI as ConnectButton/StatusChip
```

```
CORE->>CB: status_json (Connected masque-h3 direct 23ms)
```

```
CB->>BE: QMetaObject::invokeMethod(QueuedConnection) %%
```

cross-thread (§6.3)

```
BE->>EC2: sink.success(statusJson)
```

```
EC2-->>AHC: receiveBroadcastStream() event
```

```
AHC->>AHC: TunnelStatusJson.decode -> mirrored TunnelStatus
```

```
AHC-->>UI: statusStream() -> Riverpod fold -> green + "masque-  
h3 · direct · 23ms"
```

This is the Aurora analogue of doc 03 §3.3's core → FFI → UI flow — the only difference is the EventChannel-over-Frifle hop replacing frb's StreamSink. The **UI is still a pure function of the status stream** (CI2): the shim never lets the UI believe its own intent.

10. Error taxonomy + edge cases

10.1 Error mapping (C-ABI / OS → PlatformTunnelEvent / PlatformException)

Origin	Condition	Surfaced as	Kind
Sailjail/cap	tun/ CAP_NET_ADMIN denied	permissionDenied event + PlatformException("permissionDenied")	permission
open("/dev/net/tun")	EPERM/ENOENT/ EBUSY	error event, honest detail=strerror	error
ConnmanGuard.apply	route/DNS/nft install failed	error event detail="route/dns apply failed"	error
helix_core_start	HX_ERR_CONFIG/ HX_ERR_ALREADY	error; teardown (O4)	error
helix_core_attach_tun	HX_ERR_TUN (bad fd)	error; teardown; close fd	error
core status	Down{auth- failed} (revoke)	revoked event	revoked
ConnMan/OS	connection lost / backend killed	revoked (OS) or RAI down (crash)	revoked

`helix_core_last_error()` (§3.1) is the honest detail source — redacted, never a secret (§11.4.10). No error is labelled "transient/flaky" without captured evidence (§11.4.6); the RACE/SC tests (§13) reproduce each class N times.

10.2 Edge cases (each is a test point, §13)

#	Edge case	Required behaviour
E1	<code>startTunnel</code> twice (double-tap)	second is idempotent no-op success (O1)
E2	<code>stopTunnel</code> while <code>CreatingTun</code> (race)	cancel cleanly, no orphan tun/routes (O4); end in Down
E3	tun opened but <code>helix_core_start</code> fails	<code>teardown(false)</code> closes fd + reverts net + kill-switch stays closed (O-I11)
E4	ConnMan revokes connectivity mid-Up	revoked event; core re-dials (< 3 s flap, [v02-orch §7.1]); kill-switch closed during gap
E5	backend process killed (OOM/crash)	RAII <code>~HelixTunnelBackend</code> runs <code>teardown(false)</code> ; closed nft rules persist (O-I11); no leak
E6	IFF_NO_PI mismatch	corrupt packets; caught by E2E round-trip — never ship (§7.2)
E7	status callback fires after <code>stop</code> (in-flight)	trampoline checks <code>state_ != Idle</code> ; drop late events (no use-after-free)
E8	MTU edge: app sends > <code>effective_mtu</code>	core returns <code>Oversize</code> ; orchestrator lowers inner MTU ([v02-tt §4.8]); shim untouched
E9	omprussia embedder lacks a needed API	honest SKIP-with-reason (§11.4.3, AI4) + tracked migration item — never faked
E10	revoke during <code>teardown</code> (overlapping)	<code>teardown</code> is idempotent (<code>helix_core_stop</code> idempotent [v02-orch §2.4]); single down emit

11. Memory / size / performance budgets

Aurora is a **full Linux process** — it does **NOT** carry the iOS Network- Extension ~15 MB ceiling (the iOS-only make-or-break, [03-client §5.1, CI4]). But Aurora targets modest enterprise mobile hardware, so the backend stays lean by the AI3 discipline and the lean-core recipe.

Budget	Target	How	Source
Qt backend RSS (Qt runtime + core, plain-udp)	low-tens MB (UNVERIFIED ceiling — calibrate on device, §11.4.107(13))	lean Rust staticlib (opt-level=z+LTO+panic=abort+strip), no per-datagram heap growth ([v02-tt §9 MEM])	[03-client §10]

Budget	Target	How	Source
Qt backend RSS (masque-h3)	budget separately (QUIC buffers cost more — both transports measured)	cap QUIC flow-control windows if needed (the iOS fallback A applies as a knob, not a ceiling)	[03-client §5.1], [v02-tt §11]
App install size (RPM)	lean (< ~20 MB order, UNVERIFIED per Aurora packaging)	tree-shake icons/fonts, no bundled media, AOT Flutter (never Electron, C16)	[03-client §10]
Cold start	< 1 s order on target hardware	AOT, deferred non-critical providers	[03-client §10]
tun pump throughput	within the core's per-carrier budget ([v02-tt §11]); shim adds ~zero (it only hands the fd)	the shim does no crypto (AI2); throughput is the core's	[v02-tt §11]
setShields/ setExit round-trip	sub-100 ms (channel hop + core logic)	logic-only calls, no OS tunnel touched (O2)	§4.3

All numbers are **UNVERIFIED until measured on a real Aurora device** (§11.4.6/ §11.4.123) — recorded in the §13 MEM/BENCH evidence, never asserted. Aurora device unavailability ⇒ honest SKIP-with-reason for the device-gated rows (AI4).

12. Build, RPM packaging, signing

Per AI3 + [03-client §11, §5.6, 04_ARCH §5.7], Aurora is an **isolated enterprise SKU** on its own (local, §11.4.156 no-active-CI) runner — a fork lag must never block mainline.

flowchart LR

```

src["helix-ui (shared Dart)<br/>+ shims/aurora (Qt/C++ + Friflex plugin)"] --> ffi["build helix-ffi staticlib<br/>(Aurora ARM triple, opt-z+LT0+strip) (§3.3)"]
ffi --> hdr["cbindgen -> helix_core.h"]
hdr --> qt["compile HelixTunnelBackend (Qt/C++)<br/>link libhelix_core.a"]
qt --> fa["flutter-aurora build (omprussia fork, pinned tag)<br/>--dart-define=HELIX_TARGET=aurora"]
fa --> rpm["package signed RPM<br/>(Aurora SDK, Mos.Hub / omprussia toolchain)"]
rpm --> sign["RPM sign (§11.4.133 no unsigned privileged code)"]

```

```
sign --> rec["QA evidence: helixvpn---aurora-<feature>---<run-id>.mp4 (§11.4.155)<br/>release tag helixvpn-<ver> (§11.4.151)"]
```

- **Toolchain:** omprussia flutter SDK fork + flutter-embedder; plugins via Friflex / flutter-community-plugins (*_aurora variants) [research-flutter_ffi §4]. **Pin the exact fork tag** (R1); the Aurora runner is isolated [03-client §11].
- **Signing:** Aurora RPM is signed via the Aurora SDK; the privileged helper (P-B, §7.1) and any CAP_NET_ADMIN component are signed — no unsigned privileged code (§11.4.133).
- **Drift gate:** the cbindgen helix_core.h is regenerated + drift-checked against helix-ffi in the same build [03-client §11] — the Aurora backend can never silently diverge from the Rust core (a drift is a build-blocking finding, the shared-vector corpus §3.2).
- **No active CI** (§11.4.156): the "matrix" is the **local** melos + Aurora-SDK build ritual + the operator's pre-tag sweep (§11.4.40); any preserved workflow is *.disabled-local-only.

13. Test points — tied to the §11.4.169 test-type vocabulary

Every Aurora-shim workable item declares its required test types from the **§11.4.169 closed test-type vocabulary** (the LIVE constitution anchor mandating comprehensive test-type coverage); the **only** permitted absence of a warranted type is an honest §11.4.3 **SKIP-with-reason**, never a silent gap. Four-layer enforcement per §11.4.4(b) applies to every closure. The Aurora hard truth (R5): **device-gated rows are SKIP-with-reason** until an Aurora device/emulator is in the lab — this is anti-bluff honesty (AI4, §11.4.52), not a faked PASS.

Code (§11.4.169)	Type	Concrete Aurora-shim test point	Evidence (§11.4.5/.69/.107)	Device-gated?
UT	unit	C-ABI [?] frb test-vector corpus (§3.2) round-trips bit-for-bit; TunnelStatusJson.decode maps every variant (§9.1); error-code → event mapping (§10.1); openTun flag assembly (IFF_TUN IFF_NO_PI)	cargo test + Dart test logs + vector diff	no
WT	widget	shared ConnectButton/StatusChip render across every TunnelStatus fed by a fake AuroraHelixCore (override per [research-flutter_ffi §6])	golden frames	no
GOLD	golden	design system stable across themes/breakpoints (the Aurora flavor uses the same	golden PNGs	no

Code (§11.4.169)	Type	Concrete Aurora-shim test point	Evidence (§11.4.5/.69/.107)	Device-gated?
IT	integration (real System)	helix_design, so this is the shared suite) HelixTunnelBackend opens a real tun in a Linux netns standing in for Aurora; attach_tun + core pump; ping over overlay	netns capture; round-trip log	partial (netns proxy)
E2E	end-to-end	full journey on a real Aurora device: launcher → ConnectButton → tun up → curl to LAN host → StatusChip reads masque-h3 · direct · 23ms confirmed by OCR/vision (§11.4.159)	window-scoped MP4 (§11.4.155)	YES → SKIP-with-reason if no device (§11.4.3)
FA	full-automation (§11.4.25/.52/.98, deterministic §11.4.50)	N=3 identical scripted connect→status→disconnect runs producing the same ordered TunnelStatus trace over the EventChannel	3× identical status-trace artifacts	partial (netns proxy yes; device SKIP)
CH	Challenges (§11.4.27(B))	per-gate Challenge scores the captured connect-journey evidence, not config	Challenge result.json	inherits E2E gate
HQA	HelixQA	autonomous QA session drives connect→shields→exit over the channels (netns proxy)	HelixQA session evidence	partial
SEC	security (§11.4.10)	plant sessionOrMapToken value → prove it appears in no channel log / tracked file / helix_core_last_error; verify no SecretBytes-class leak; Sailjail profile least-privilege audit	grep-empty proof; profile audit	no
SC	stress + chaos (§11.4.85)	iface flap (ConnMan down/up) → revoked + core < 3 s recovery; backend process-kill mid-transfer → RAII teardown leaves no leak (E5); stopTunnel during CreatingTun (E2)	recovery trace; leak-scan after kill	partial (netns yes; device SKIP)
CONC	concurrency / atomicity	concurrent setShields (channel) + status callback (tokio thread) → no torn state_; double startTunnel (E1)	concurrency harness log	no

Code (§11.4.169)	Type	Concrete Aurora-shim test point	Evidence (§11.4.5/.69/.107)	Device-gated?
RACE	race / deadlock	status callback after stop (E7); cross-thread sink touch detection (must use QueuedConnection, §6.3); --features loom on the C-ABI state cell	loom / TSan report	no
MEM	memory	Qt-backend RSS for plain-udp and masque (no per-datagram heap growth); no fd leak across 100 connect/disconnect cycles	/proc/<pid> RSS sample; fd-count CSV	partial (device for true target RSS → SKIP)
BENCH	benchmarking	tun-pump throughput (shim adds ~0); setShields/setExit round-trip latency; effective_mtu measured not assumed ([v02-tt §11])	throughput/latency CSV	partial

13.1 Anti-bluff rule for this shim (§11.4.5/.69/.107)

Every Aurora-shim PASS ships **captured evidence** (netns round-trip log, device MP4, RSS CSV, grep-empty secret proof), never a config-only or grep-only PASS. A green widget test does **not** prove the Aurora user is protected — the IT/E2E/SC layers are the proof, and the user-visible claim is earned only by the device recording. Where the device is genuinely absent, the row is an honest **SKIP-with-reason** (hardware_not_present per §11.4.69 / §11.4.3) with a tracked migration item — **the Phase-3 biggest-platform-risk discipline made mechanical** (AI4, [SYN §4]).

14. Open decisions surfaced by this document

Per §11.4.6/§11.4.66 — options + recommendation, never silently resolved. These are *Aurora-shim* decisions (the program-level D1–D8 in doc 00, the client-stack D-CLIENT-1..5 in [03-client §14]).

#	Decision	Option A	Option B	Recommendation
D-AUR-1	Where the core lives on Aurora	Qt/C++ backend via cbindgen C ABI (this spec, §2)	core in the Flutter process via frb (like Linux)	A — A depends only on a universal C ABI + Friflex (both proven); B depends on frb-in-the-omprussia-embedder which is UNVERIFIED (R2). Gate B on a frb-on-fork spike; if it passes, B collapses the status plumbing to frb and is simpler. Surface to operator.

#	Decision	Option A	Option B	Recommendation
D-AUR-2	Backend process model	in-process with the embedder (fd shared directly)	separate D-Bus service (fd via SCM_RIGHTS, §7.4)	Gate on the §7.1 permission spike — if Sailjail permits in-process tun (P-A), prefer in-process; if not, the privileged-helper service (P-B) forces the separate-service model. Real platform measurement, not a guess (§11.4.6).
D-AUR-3	VPN permission path	P-A in-process CAP_NET_ADMIN	P-B privileged D-Bus helper daemon	Measure first (§7.1). Prefer P-A (simpler); fall to P-B (Windows-service-analogue [03-client §5.3]) if Sailjail denies. Honest SKIP if neither (AI4).
D-AUR-4	Route/DNS mechanism	ConnMan D-Bus native	direct rtnetlink + nftables + /etc/resolv.conf	Prefer ConnMan-native (avoids fighting ConnMan's DNS manager) if its D-Bus surface supports it (UNVERIFIED, §7.3); netlink/nft is the proven fallback.
D-AUR-5	Status bridge granularity	two EventChannels (lifecycle helixvpn/tunnel/events + status helixvpn/core/status, this spec)	one merged channel	A (two channels) — keeps TunnelPlatform.events() = lifecycle-only per the [03-client §4] contract (shared with every other shim) and the status bridge Aurora-specific; cleaner separation, no merged-payload demux.

15. Cross-document contracts this document fixes

Contract	Fixed value	Consumed by
Aurora cbindgen C ABI (helix_core_* in helix_core.h, §3.1) + the frb-correspondence (§3.2)	§3	helix-ffi (must emit the C ABI), the Qt backend
Friflex channel contract (helixvpn/tunnel, helixvpn/tunnel/events, helixvpn/core, helixvpn/core/status + payload schemas, §4)	§4	the Friflex plugin, AuroraTunnelPlatform, AuroraHelixCore
AuroraTunnelPlatform/AuroraHelixCore channel-backed impls of the existing abstractions (AI1)	§5	helix_domain wiring (the one Aurora-aware line, §5.3)
	§9	the FFI contract test [03-client §12]; every Aurora widget

Contract	Fixed value	Consumed by
Aurora status JSON [?] TunnelStatus mapping (consistency with [v02-orch §4.1] + [03-client §3.1])		
tun fd / IFF_NO_PI / kill-switch-on-crash Aurora mechanics	§6–§8	the Qt backend, the device smoke test
Phase-3 risk + honest-SKIP discipline (device-gated rows SKIP-with-reason)	§13, AI4	the coverage ledger [03-client §12]; release-gate sweep

Sources verified

- `final/03-client-core-and-ui.md` [03-client] — §3 (FFI surface + TunnelStatus mirror + Shields/ExitOption), §3.2 (HelixCore Dart abstraction), §3.3 (FFI $\rightleftharpoons$ shim seam), §4 (TunnelPlatform contract + obligations O1–O5 + TunnelConfig
 - PlatformTunnelEvent), §5/§5.6 (per-platform matrix + Aurora row: Qt/C++ + tun, core as C lib, Friflex), §6 (flavor/capability wiring), §7 (design system, branch-on-size-not-Platform), §8 (Riverpod pure-function-of-status), §10 (budgets), §11 (build/sign/drift, no-CI), §12 (testing + FFI contract test), §13 (Phase-3 T3.2 Aurora), §14 (D-CLIENT-1..5, incl. D-CLIENT-3 cbindgen-for-Aurora).
- `final/v02-data-plane/orchestrator-and-state.md` [v02-orch] — §2.4 (graceful teardown order), §4.1 (frozen TunnelStatus enum — the consistency anchor), §5.1 (OrchState + Idle/ShuttingDown projection), §7.1 (< 3 s roam recovery), §8 (kill-switch + DNS coupling; §8.2 arm-closed, §8.4 closed-on-crash O-I11, §8.6 open-only-on-user-stop).
- `final/v02-data-plane/transport-trait.md` [v02-tt] — I1 (transport never sees plaintext), §4.8 (effective_mtu/oversize), §8.3 (shared encode/decode test-vector corpus pattern), §9 (§11.4.169 test-type vocabulary), §11 (perf budget).
- `04_VPN_CLD/HelixVPN-Architecture-Refined.md` [04_ARCH] — §5.2 (why Flutter; Aurora OMP fork the only path; KMP not viable), §5.3 (shim matrix incl. Aurora Qt/C++ + tun + C lib), §5.7 (Aurora enterprise SKU, Russian-hosted toolchain, own CI/signing).
- `04_VPN_CLD/HelixVPN-helix-ui-Flutter.md` [04_UI] — §6/§6.1 (Aurora row: Qt/C++ backend, core as C lib, Friflex bridge, OMP fork; Russian/Chinese first-tier I10n), §11 (CI matrix: separate isolated Aurora runner, RPM signing).
- `v09-research/research-flutter_ffi.md` [research-flutter_ffi] — §2/§4 (UniFFI-Dart not production-ready $\rightarrow$ raw C for Aurora; omprussia flutter+flutter-embedder+ flutter-community-plugins, *_aurora pattern, Mobius 2025, second-tier target, bespoke per-platform VPN plugin, honest SKIP where embedder lacks API), §3 (staticlib/cdylib packaging), §6 (Riverpod StreamProvider + overrideWithValue for fakes), §7 (synthesis).
- `v09-research/_SYNTHESIS.md` [SYN] — §4 (Phase-3 Aurora real-native-work, biggest platform risk), §5 (client architecture, shim is the only platform code), §9 (constitution bindings: §11.4.151/.155 prefixes, §11.4.156 no-CI, §11.4.133).

Constitution: §11.4.44 (revision header), §11.4.6/§11.4.66 (decisions = options + recommendation; UNVERIFIED never fabricated), §11.4.3/§11.4.52 (honest SKIP-with-reason, no faked device PASS), §11.4.5/§11.4.69/§11.4.107/§11.4.159 (captured/window-scoped evidence), §11.4.10 (no secret leak), §11.4.27/§11.4.50 (no-fakes + deterministic), §11.4.151/§11.4.155 (release/recording prefixes), §11.4.156 (no active CI), §11.4.133 (signed privileged code), §11.4.169 (test-type coverage vocabulary).

End of nano-detail specification — Aurora OS shim (Qt/C++ tun) (Volume 4, Clients). Deepens [03-client §5.6]; pairs with [v02-orch §4.1] (the TunnelStatus consistency anchor) and the sibling shim docs. Surfaced decisions: D-AUR-1 (core-in-Qt-backend vs frb), D-AUR-2 (process model), D-AUR-3 (VPN permission path), D-AUR-4 (route/DNS mechanism), D-AUR-5 (status-bridge granularity) — all presented, none silently resolved (§11.4.66).